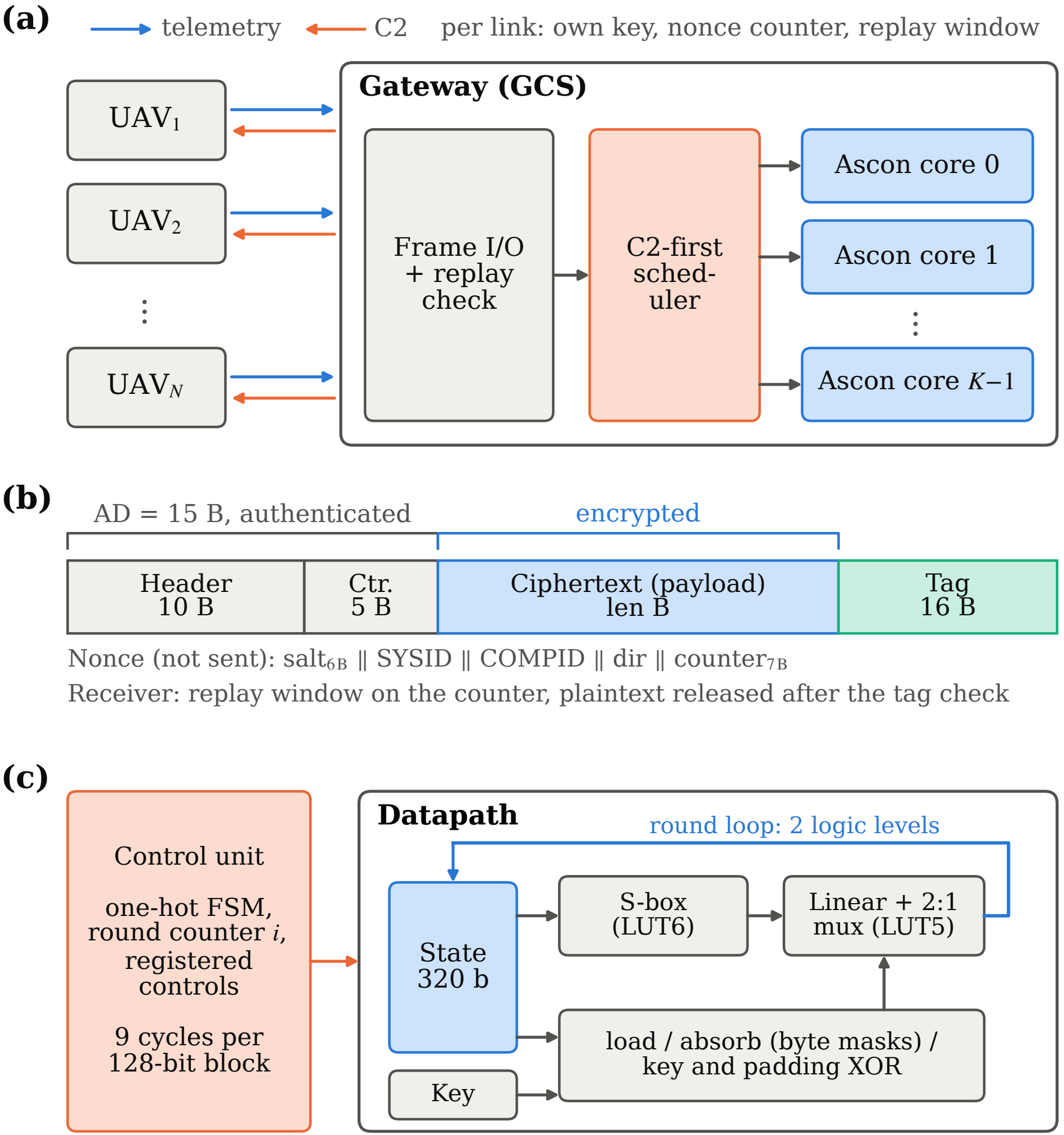
(a)
telemetry
C2
per link: own key, nonce counter, replay window
UAV1
UAV2
UAVN
Gateway (GCS)
Frame I/O + replay check
C2-first sched-uler
Ascon core 0
Ascon core 1
Ascon core K−1
(b)
AD = 15 B, authenticated
encrypted
Header 10 B
Ctr. 5 B
Ciphertext (payload) len B
Tag 16 B
Nonce (not sent): salt6B ‖ SYSID ‖ COMPID ‖ dir ‖ counter7B
Receiver: replay window on the counter, plaintext released after the tag check
(c)
Control unit
one-hot FSM, round counter i, registered controls
9 cycles per 128-bit block
Datapath
round loop: 2 logic levels
State 320 b
S-box (LUT6)
Linear + 2:1 mux (LUT5)
load / absorb (byte masks) / key and padding XOR
Key